# Heatmap and Vector Field Visualization of Whole-Slide Images in Digital Pathology

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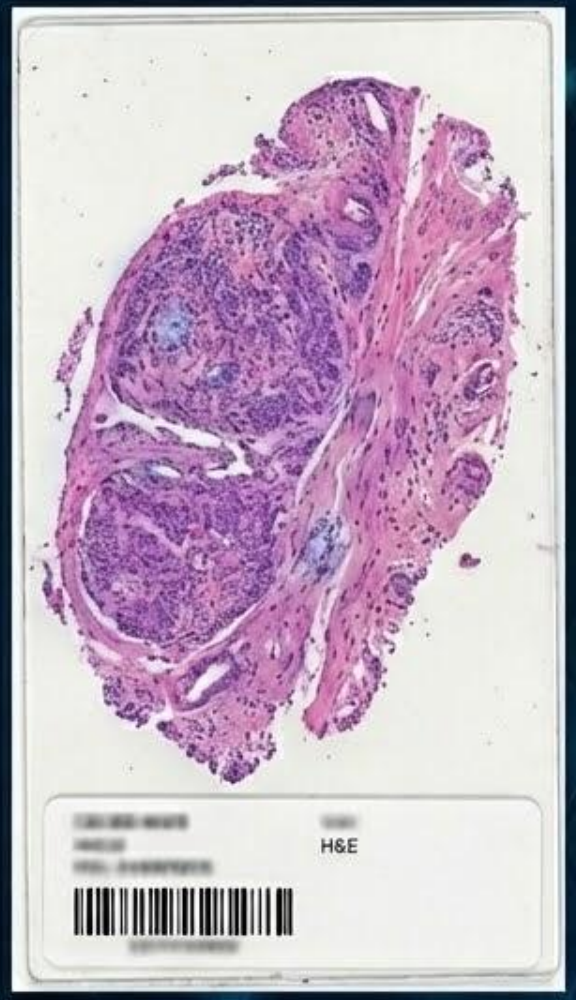
# Explainable AI

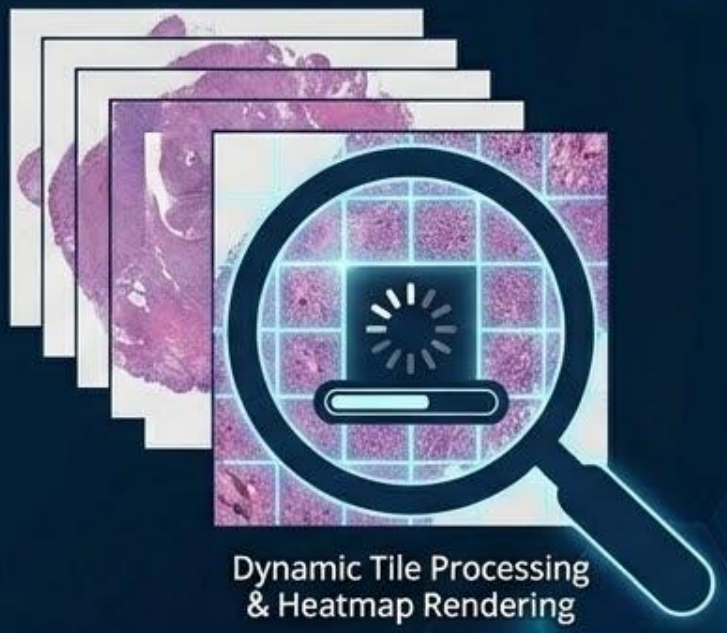
Source: www.nature.com/scientificreports/

- Deep learning models can predict recurrence risk from histopathology images.
- Patch-based models lose global spatial context and are difficult to interpret.
- AI models can analyze these pathology images quickly and accurately, but Whole Slide Images are extremely large and complex.
- Even if AI is fast and accurate, pathologists need quick and clear understanding to make treatment decisions.
- Explainable AI in pathology is therefore crucial, we need to show the results of AI the best way.



# Difficulty in Visualizing and Processing Whole-Slide Images

- Whole Slide Images are multi-resolution, multi-level, and extremely large.
- They cannot be visualized efficiently using standard web frameworks.
- Heatmap overlays require precise spatial calibration across zoom levels.
- Zooming in/out requires dynamic tile processing and real-time heatmap rendering.
- Static visualization methods are not sufficient for WSI-scale applications.



# Proposed Framework/Tool

## Multi-Model Architecture:

**CellViT**
Vision Transformers for nuclei classification

**HoverNeXt**
Dense instance segmentation

**SAM (Segment Anything Model)**
Zero-shot generalization

→ **Integrated System Core**

## User-Adaptive Processing:

**Pathologist**

→ **Clinical Question**
e.g., Focus on lymphocytes, tumor, or stroma

↓ **Optimal Model Selection**

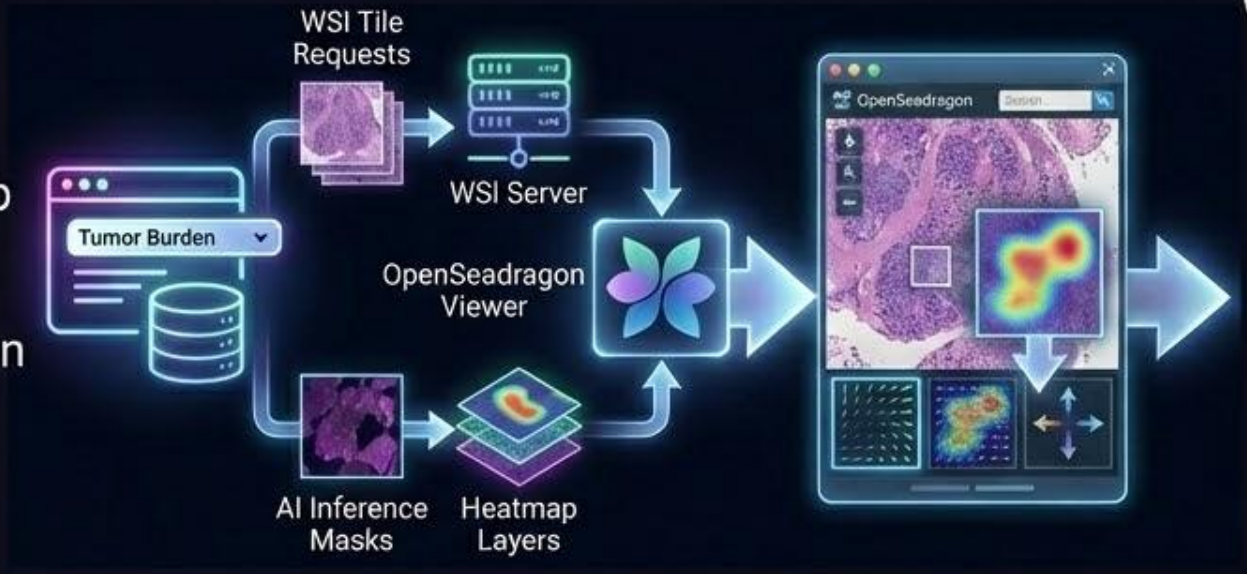
↓ **Visualize Results**

→

# Heatmap Overlay with Openseadragon

## Interactive Web Interface

- Built on a Python/Flask (or Django/FastAPI) web server.
- User Controls: Dropdown menus for specific cell targets (e.g., 'Highlight Tumor Burden').



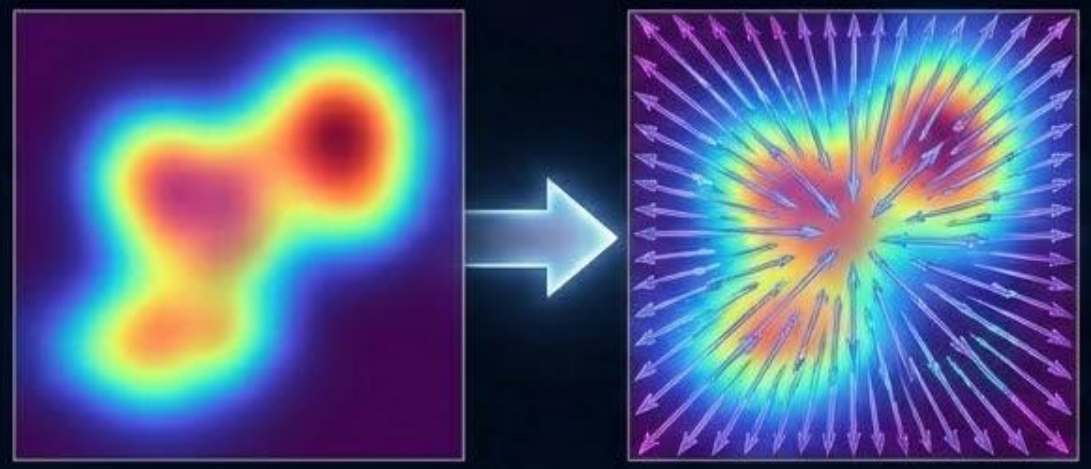

## Dynamic Visualization with OpenSeadragon

- Real-time Tiling: Fetches only visible tiles.
- Overlay Logic: AI masks convert to heatmap layers.
- Flow Visualization: Overlays vector fields for directionality.

# Dynamic Flow & Spread Visualization

## Beyond Static Heatmaps & Tile-Based Vector Fields

- **Beyond Static Heatmaps**: Standard heatmaps show density, Flow Diagrams show directionality.
- **Tile-Based Vector Fields**: Flow direction calculated for each WSI tile.
- Generates directional vectors (arrows) for spread path.

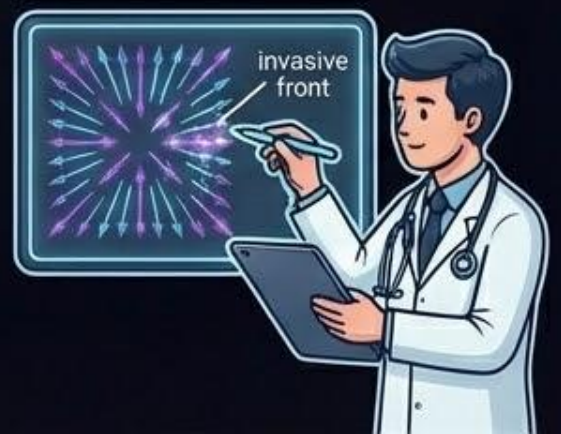


**Static Heatmap (Density)**

**Flow Diagram (Directionality)**

## Clinical Insight

- Helps pathologists identify invasive fronts and aggressive growth patterns instantly.
- Provides critical context for writing clinical treatments.



# Q & A?

# Thank You!